

Ahmad Raza

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SUMMARY

Software Engineering Graduate specializing in AI, Machine Learning, and NLP with hands-on experience building multi-agent LLM systems. Research focus on retrieval-augmented generation and cognitive adaptation in educational systems. Adept at driving innovation and generating measurable improvements across diverse projects.

EDUCATION

Thal University Bhakkar

Bachelor of Science in Software Engineering

Punjab

Graduated May 2026

- CGPA: 3.86/4.0
- Relevant Coursework: Software Engineering, Data Structures & Algorithms, Web Engineering, Database Systems, Cloud Computing

PROJECTS

Sage (FYP) | *LangChain, LangGraph, llama.cpp, FastAPI, React* | [GitHub](#)

Feb–May. 2026

- Built an offline multi-agent academic assistant (8 LangGraph agents) with a dual-model inference; hybrid RAG pipeline (teammate-led) combining dense retrieval and BM25.
- Engineered a hardware-aware llama.cpp lifecycle manager with dynamic context-window scaling (3k–64k), VRAM-based GPU offloading, and reduced server cold-start from 17s to 4s via BayesOpt-tuned compiler flags.
- Shipped as a Windows desktop app across 4 tiered builds (CPU/CUDA x full/lite) with SHA-256 integrity validation on all staged artifacts; CI/CD via GitHub Actions with automated builds upload to Cloudflare R2.
- Enabled zero-code institutional adoption via a TOML-config driven end-to-end pipelines, featuring fully offline, privacy-first deployment in resource-constrained environments.

Seq Modeling From Scratch | *Python, NumPy, Math* | [GitHub](#)

Nov. 2025

- Implemented RNN, LSTM, GRU without auto-differentiation frameworks to master gating mechanisms and computational graphs.
- Manually derived and coded forward and backward passes (Backpropagation Through Time) to manage gradient flow in dynamic networks and enable training.
- Built Seq2Seq models with Bahdanau and Luong attention mechanisms specifically for language modeling tasks. Trained character and word-level language models to demonstrate practical sequence learning.

NeuroScope | *Python, NumPy, Matplotlib* | [GitHub](#) | [PyPI](#) | [Docs](#)

Sep 2025

- Engineered a production-grade Python framework for informed training and diagnosis of neural networks.
- Developed a real-time diagnostics system monitoring 10 health indicators (e.g., Dead Neurons, Gradient SNR, Weight Update Ratios) and a comprehensive analytics and evaluation system validated against research.
- Designed dynamic visualizations for neuron activation, gradients, and weights histograms and evolution plots to track training dynamics.
- Maintained 60%+ test coverage and automated PyPI publishing workflows via GitHub Actions.

RESEARCH

CLAG: Stateless Cognitive Adaptation for Retrieval-Augmented Generation | *In Preparation*

2026

- Proposed a stateless cognitive adaptation framework for educational RAG that estimates a learner's cognitive level from a single query via Bayesian posterior fusion with EM-estimated population priors.
- Derived and empirically validated theoretical guarantees, confirming single-query adaptation viability and strict adaptation monotonicity.
- Demonstrated via a comprehensive evaluation pipeline that the Bayesian formulation achieves highly accurate Zone of Proximal Development (ZPD) targeting ($\text{MAE} \approx 0.068$) and significantly outperforms linear baselines ($p < 0.001$).

TECHNICAL SKILLS

Deep Learning & ML: LLMs, Transformers, Fine-tuning, RAG, Multi-Agent Systems, Sequence Models, Representation Learning, Statistical Learning Theory, Self-Supervised Learning

Foundations: Linear Algebra, Multivariate Calculus, Probability Theory, Statistics, Optimization

MLOps & Infra: FastAPI, MLflow, DVC, Model Deployment & Serving, Docker, CI/CD

Languages & Libraries: Python, SQL, NumPy, Matplotlib, Seaborn, Scikit-Learn, XGBoost, Optuna, PyTorch, LangChain, LangGraph

CERTIFICATIONS

DeepLearning.AI (Coursera): Deep Learning Specialization, Machine Learning Specialization, Mathematics for Machine Learning & Data Science Specialization

Other Specializations: MLOps Specialization (Duke University), IBM Machine Learning Professional Certificate